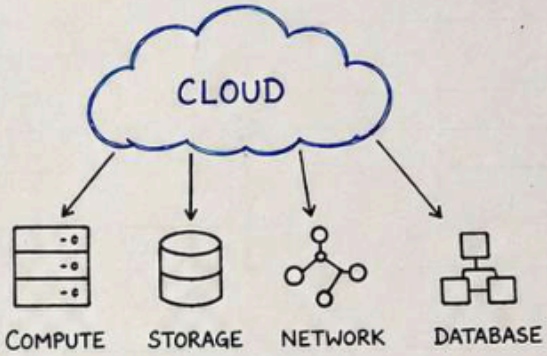
@aman_views

# CLOUD COMPUTING

## — NOTES —

### 1) What is Cloud Computing?

Cloud Computing is the delivery of computing services — servers, storage, databases, networking, software, analytics, intelligence — over the internet ("the cloud") to offer faster innovation, flexible resources and economies of scale.



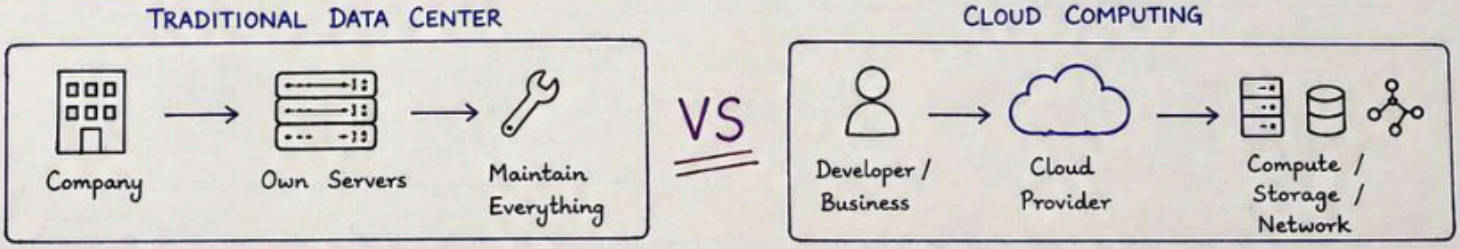

### 2) Why do we need Cloud?

- No need to buy or maintain expensive hardware.
- Scale resources up or down based on demand.
- Pay only for what you use.
- High availability, reliability and global reach.
- Focus on building applications, not managing infra.

**Think of Cloud as...**

Electricity from a plug point instead of generating your own power plant.

### 3) Traditional Data Center vs Cloud




- ✗ High upfront cost
- ✗ Need IT team for maintenance
- ✗ Limited scalability
- ✗ Downtime risk is higher

- ✓ Low cost, pay-as-you-go
- ✓ No hardware maintenance
- ✓ Highly scalable & flexible
- ✓ High availability & reliability

### 4) Key Characteristics of Cloud

**On-demand**
Resources are available whenever you need them.

**Scalability**
Scale up or down instantly based on your demand.

**Elasticity**
Resources automatically adjust to handle traffic changes.

**Pay-as-you-go**
You pay only for what you use.

**Global Availability**
Deploy across multiple regions and reach users globally.

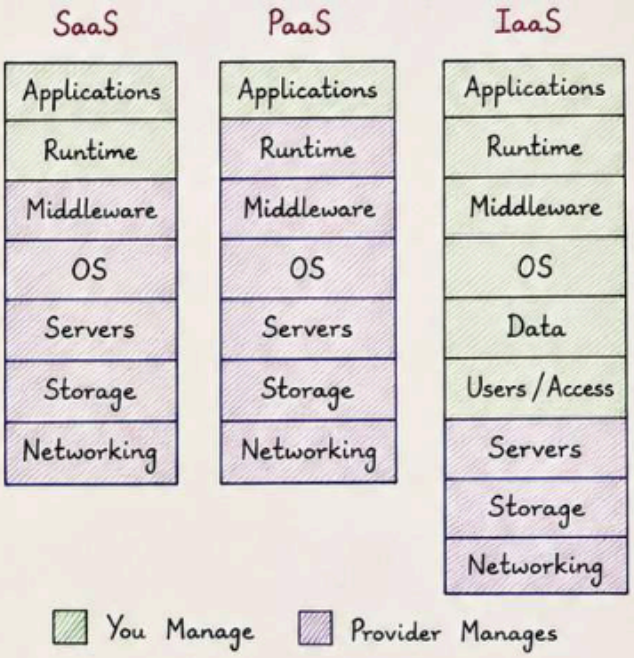
**Real-world Example :** Netflix scales to millions of users worldwide during releases without worrying about servers. They use cloud!

# 2) Cloud Service Models

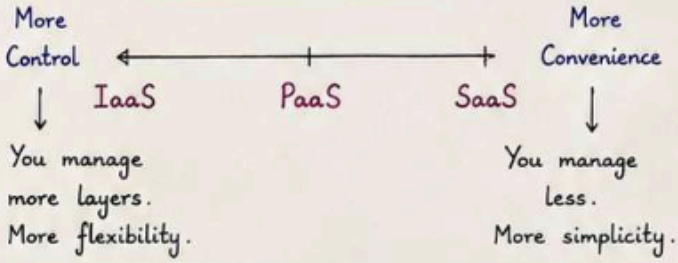
Cloud providers deliver services in different layers. You choose how much control you want and they mangle.

| Model | You Manage | Provider Manages |
|---|---|---|
| SaaS (Software as a Service) | • Data<br>• Users / Access | • Applications<br>• Runtime<br>• Middleware<br>• OS<br>• Servers<br>• Storage<br>• Networking |
| PaaS (Platform as a Service) | • Applications<br>• Data<br>• Users / Access | • Runtime<br>• Middleware<br>• OS<br>• Servers<br>• Storage<br>• Networking |
| IaaS (Infrastructure as a Service) | • Applications<br>• Runtime<br>• Middleware<br>• OS<br>• Data<br>• Users / Access | • Servers<br>• Storage<br>• Networking |

## How They Differ (Layer View)




## Control vs Convenience




## Examples

| | |
|---|---|
| SaaS → | Gmail, Google Workspace, Microsoft 365, Salesforce |
| PaaS → | Google App Engine, Heroku, Azure App Service |
| IaaS → | AWS EC2, Azure Virtual Machines, Google Compute Engine |

## Key Takeaways

- → Choose the model based on your needs, skills and business goals.
- → More control = More responsibility.
- → More convenience = Less control and more abstraction.

## When to Use What ?

| | |
|---|---|
| SaaS → | When you just want to use software and don't want to worry about infrastructure. |
| PaaS → | When you want to focus on building and deploying your application, not managing servers. |
| IaaS → | When you need full control over infrastructure and have specific requirements. |

**Pro Tip**

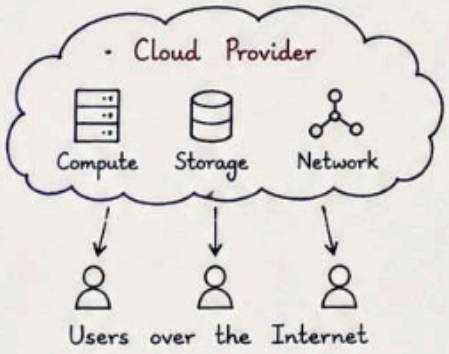
Understand these models clearly. This is one of the most asked interview topics!

# 3) Deployment Models

Cloud can be deployed in different ways depending on the organization's needs, security, and control.

## 1. Public Cloud

- Owned and managed by a cloud provider.
- Resources are shared over the internet.
- Highly scalable and cost-effective.
- Examples : AWS, Azure, GCP

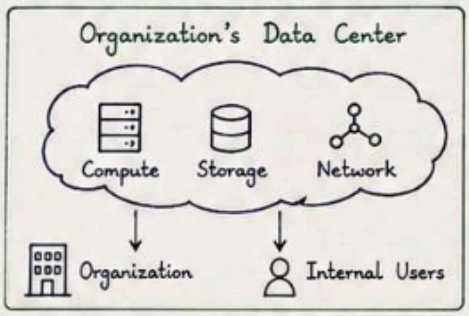



**When to use ?**

When you need high scalability, global reach and pay-as-you-go model.

## 2. Private Cloud

- Cloud infrastructure used exclusively by one organization.
- More control, security and customization.
- Higher cost.

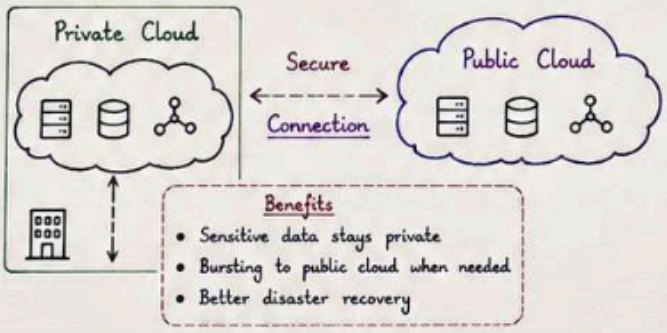



**When to use ?**

When you have strict security, compliance or regulatory requirements.

## 3. Hybrid Cloud

- Combination of public cloud and private cloud.
- Data and apps move between both.
- Gives flexibility and more control.

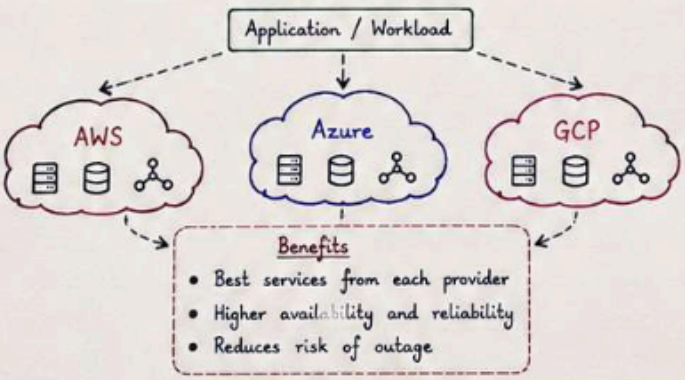



**When to use ?**

When you want to keep some workloads private and use public cloud for flexibility and scale.

## 4. Multi-Cloud

- Using services from multiple cloud providers.
- Avoids vendor lock-in.
- Increases availability and flexibility.




**When to use ?**

When you need best-in-class services from different providers or want to avoid dependence on a single cloud.

**Key Point :** Choose the right deployment model based on your business needs, control, security, compliance and cost. ---> There is no "one-size-fits-all" in cloud.

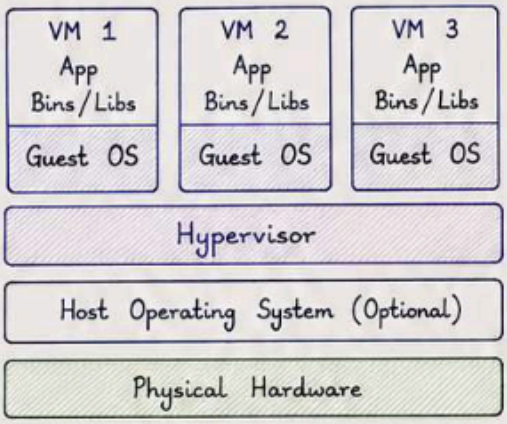


# 4) Virtualization & Containers

Virtualization and containers help us run multiple applications efficiently on the same infrastructure.

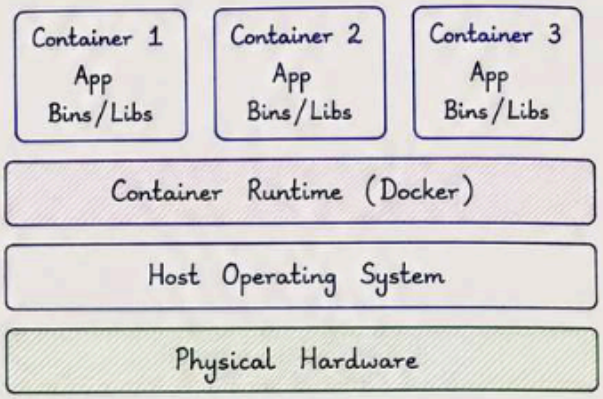
## A. Virtual Machines (VM)

- A VM is a virtual computer that behaves like a physical machine.
- Each VM has its own OS.
- Heavier but more isolated.




## B. Containers

- Containers package the application and its dependencies together.
- They share the host OS kernel.
- Lightweight, portable and fast.




## C. VM vs Container

| Feature | Virtual Machine | Container |
|---|---|---|
| Isolation | Strong isolation (each VM has own OS) | Process-level isolation |
| Size | Large (GBs) | Small (MBs) |
| Boot Time | Slow (minutes) | Fast (seconds) |
| Resource Usage | High | Low |
| Portability | Medium | High |
| Best Use | Legacy apps, different OS | Microservices, modern apps |

## D. Docker (Containerization Platform)

- Docker helps build, ship and run containers easily.
- Uses Dockerfile to define image.
- Images run as containers.

Code → Dockerfile → Image → Container

**Common Docker Commands**

- `docker build -t myapp .` → Build image
- `docker images` → List images
- `docker run myapp` → Run container
- `docker ps` → List running containers
- `docker stop <id>` → Stop container

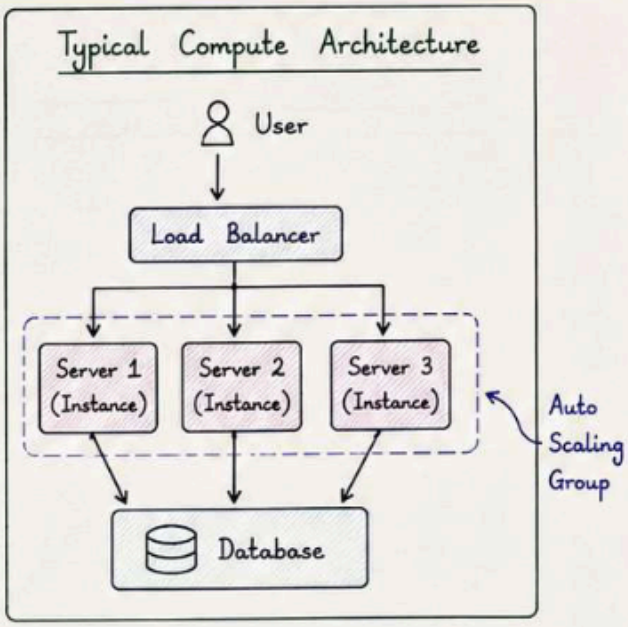
### Key Takeaways

- ✓ VMs are fully virtualized, containers are OS-level virtualization.
- ✓ Containers are lightweight and start very fast.
- ✓ Docker makes containerization simple and consistent across environments.
- ✓ Containers are ideal for microservices and cloud-native applications.

**Pro Tip**: Use containers when you need speed, scalability and portability. Use VMs when you need strong isolation or different OS.

# 5) Cloud Compute

Compute services let you run your applications in the cloud.




## A. Compute Options

1. Virtual Machines (VMs) → Full control over OS, apps.
2. Compute Instances → Pre-configured VM in cloud.
3. Containers → Lightweight, portable applications.
4. Serverless → Run code without managing servers.
5. Auto Scaling → Automatically scale based on demand.

## B. Scaling

### Scale Up (Vertical Scaling)

- Increase the power of a single machine
- Example: Increase CPU / RAM



### Scale Out (Horizontal Scaling)

- Add more machines to distribute load
- Example: Add more instances



### Note

Most cloud apps use Scale Out for better availability and fault tolerance.

## C. Serverless (Functions as a Service)

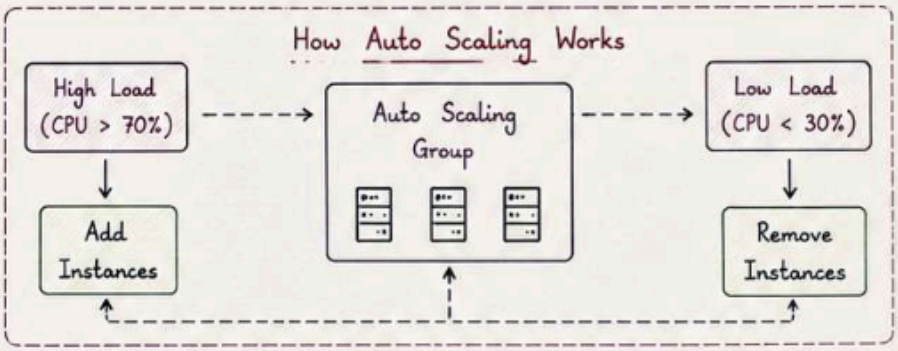
Event / Request (HTTP, S3, etc.) → Cloud Function (Your Code) → Services (DB, Storage, Queue, etc.) → Response / Result

### Benefits

- No server management
- Pay only for execution time
- Auto scale to zero
- Highly available by default

## D. Auto Scaling

- Automatically adjusts the number of instances based on demand.
- Uses rules based on metrics like CPU utilization, Memory, Requests.
- Helps in performance and cost optimization.





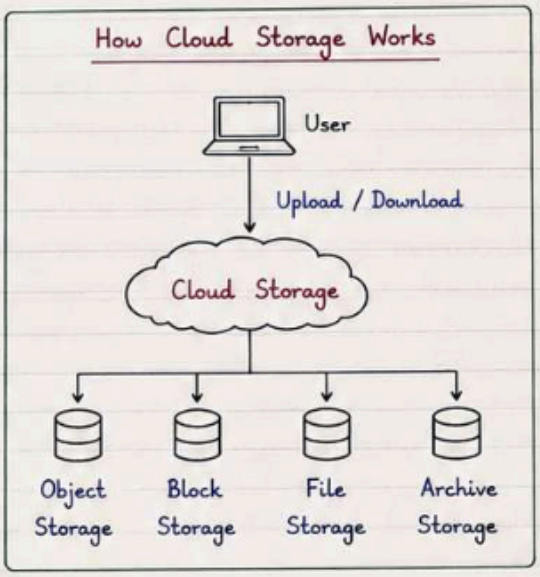

**Pro Tip:** Design for scale from day 1. Use Auto Scaling, Load Balancing and Stateless applications to handle growth smoothly.

# 6) Storage in Cloud

Cloud storage allows you to store and access data from anywhere with high durability and availability.

## A. Types of Cloud Storage

| Type | Use | Examples |
|---|---|---|
| Object Storage | • Store large amounts of unstructured data<br>• Access via API | AWS S3, Azure Blob<br>Google Cloud Storage |
| Block Storage | • For databases, VMs<br>• Low latency, high performance | AWS EBS, Azure Disk<br>Google Persistent Disk |
| File Storage | • Shared file systems<br>• Used in lift-and-shift apps | AWS EFS, Azure Files<br>Google Filestore |
| Archive Storage | • Long-term backup<br>• Infrequent access | AWS Glacier, Azure Archive<br>Google Nearline |




## B. Key Benefits

- Durable – Data is replicated across multiple locations.
- Available – High availability with minimal downtime.
- Secure – Encryption at rest and in transit.
- Scalable – Store from GBs to PBs without hardware.

**Pro Tip**

Choose storage type based on your data access pattern, performance needs and cost.

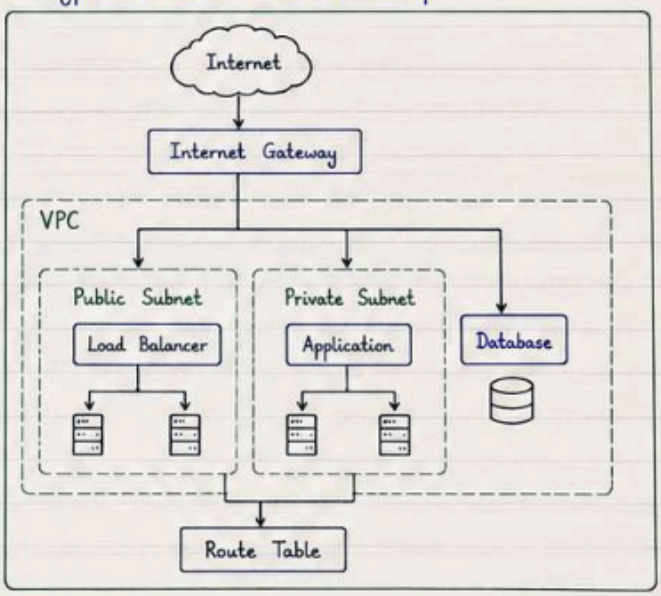
# 7) Networking in Cloud

Cloud networking connects resources securely and enables communication between users, applications and services.

## A. Key Networking Components

| Component | Purpose |
|---|---|
| VPC (Virtual Private Cloud) | Isolated network for your resources |
| Subnet | Segment of VPC (public / private) |
| Internet Gateway | Connects VPC to the internet |
| Route Table | Controls traffic flow in the VPC |
| Security Group | Acts as virtual firewall (stateful) |
| Load Balancer | Distributes traffic across instances |

## B. Typical Cloud Network Setup




**Pro Tip :** Use private subnets for backend resources like databases.
Use security groups to allow only required traffic.

# 8) Databases in Cloud

Cloud databases are managed services that provide scalability, high availability and security.

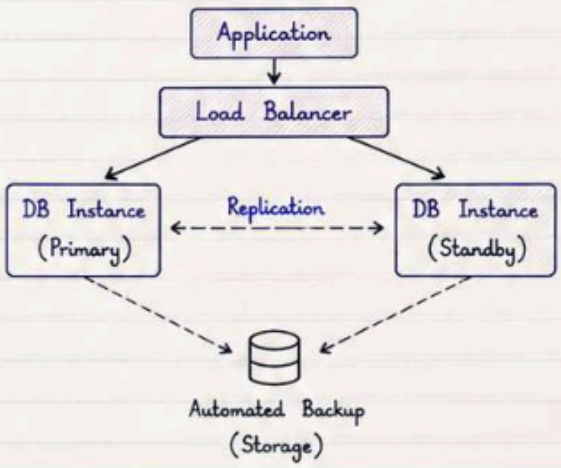
## A. Types of Cloud Databases

| Type | Description | Examples |
|---|---|---|
| Relational (SQL) | Structured data in tables with schema. | Amazon RDS, Cloud SQL, Azure SQL |
| NoSQL | Flexible schema, key-value, document, wide-column. | DynamoDB, MongoDB Atlas, Cassandra |
| In-Memory | Stores data in memory for ultra-fast access. | Redis, Memorystore |
| Data Warehouse | Designed for analytics and large scale queries. | Amazon Redshift, BigQuery, Snowflake |
| Graph Database | Stores data as nodes and relationships. | Amazon Neptune, Neo4j Aura |

## B. Benefits

- Automated backups and disaster recovery.
- Easy scaling (up or down).
- High availability with minimal downtime.
- Reduced operational overhead.

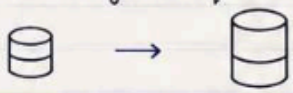
### High Availability Setup (Example)




## C. Scaling Options

### Vertical Scaling (Scale Up)

- Increase power of existing DB instance.
- More CPU, RAM, IOPS.
- Downtime may be required.



### Horizontal Scaling (Scale Out)

- Add more instances and distribute load.
- Read replicas, sharding.
- Better for high availability.



### Pro Tip

Use managed database services to focus on your application, not on database administration.

## D. Security Best Practices

1. Enable encryption at rest and in transit.
2. Use IAM roles for access control.
3. Enable Multi-AZ deployment.
4. Regular backups and point-in-time recovery.
5. Monitor with alerts and audit logs.

## E. Common Cloud Database Services

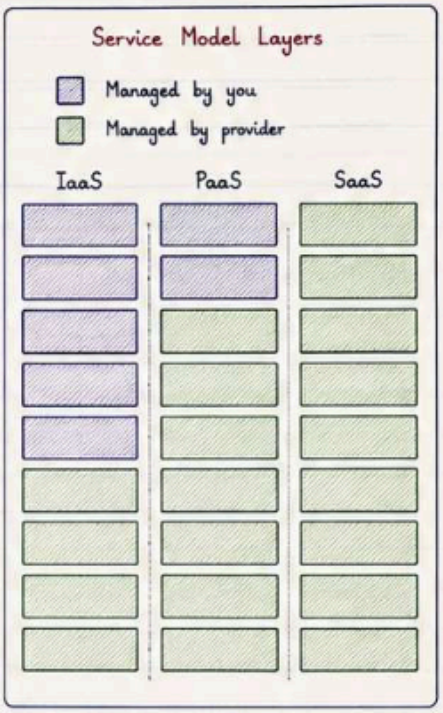
| Provider | Relational (SQL) | NoSQL | Data Warehouse |
|---|---|---|---|
| AWS | Amazon RDS, Aurora | DynamoDB | Redshift |
| Azure | Azure SQL Database | Cosmos DB | Synapse Analytics |
| GCP | Cloud SQL (PostgreSQL, MySQL) | Firestore, Bigtable | BigQuery |

**Pro Tip :** Choose the right database type based on your data model, access pattern and consistency requirements.

## 9) Cloud Services Models

Cloud services models define the level of control, management and flexibility provided by the cloud provider.

| Model | What You Get | You Manage | Best For |
|---|---|---|---|
| IaaS (Infrastructure as a Service) | • Virtual machines<br>• Storage<br>• Networking<br>• Operating systems | • OS, middleware<br>• Runtime<br>• Data<br>• Applications | Maximum control and flexibility. Suitable for custom environments. |
| PaaS (Platform as a Service) | • Runtime environment<br>• Middleware<br>• Development tools<br>• Database | • Applications<br>• Data<br>• Configurations | Developers who want to focus on code, not on infrastructure. |
| SaaS (Software as a Service) | • Complete applications<br>• Ready to use<br>• Accessible via browser | • Data (limited)<br>• User management (some settings) | End users who want ready-to-use software without management. |




## 10) Cloud Pricing Models

- Cloud pricing is based on pay-as-you-go.
- You only pay for what you use.
- Helps in cost optimization.

**Pay-as-you-go**
Pay for compute, storage, network etc. by actual usage.

**Reserved Instances**
Commit for 1 or 3 years and get discounts.

**Spot Instances**
Use spare capacity at lower price. Can be interrupted.

**Subscription**
Pay fixed amount regularly for services.

**Free Tier**
Limited free usage for learning and testing.

## 11) Advantages of Cloud Computing

1. Scalability – Easily scale up or down based on demand.
2. Cost Efficiency – Pay only for what you use.
3. High Availability – Built-in redundancy and disaster recovery.
4. Flexibility – Access resources from anywhere.
5. Speed – Quick provisioning and deployment.
6. Global Reach – Deploy applications closer to users.
7. Security – Advanced security features and compliance.
8. Innovation – Access to latest tools and services.

## 12) Challenges of Cloud Computing

1. Security Risks – Data breaches and unauthorized access.
2. Vendor Lock-in – Moving out from one provider is complex.
3. Downtime – Service outages can affect business.
4. Compliance – Meeting regulatory requirements.

**Pro Tip**

Start with small workloads, monitor costs, choose the right service model, and follow best practices for security and backup.

# 13) Cloud Security Best Practices

1. Follow shared responsibility model.
2. Use strong IAM policies and least privilege.
3. Enable Multi-Factor Authentication (MFA).
4. Encrypt data at rest and in transit.
5. Regularly monitor and review access.
6. Keep backups and test recovery.
7. Use security groups, firewalls and network segmentation.
8. Keep systems and services updated.
9. Enable logging and audit trails.
10. Conduct regular security assessments.

**Defense in Depth**

- Identity & Access Management
- Network Security
- Data Protection
- Monitoring & Logging
- Incident Response

**Key Security Services (Examples)**

| AWS | Azure | GCP |
|---|---|---|
| IAM | Azure AD | Cloud IAM |
| Security Hub | Defender for Cloud | Security Command Center |
| KMS | Key Vault | Cloud KMS |
| WAF | WAF | Cloud Armor |
| GuardDuty | Sentinel | Chronicle |

**Pro Tip**

Security in cloud is not a one-time setup, it's an ongoing process. Monitor, review and improve continuously.

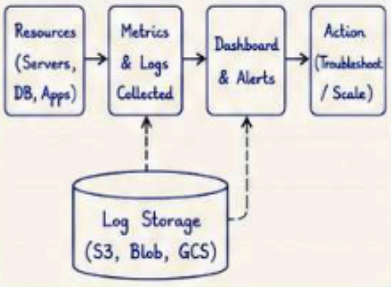
# 14) Cloud Monitoring & Logging

Monitoring ensures performance, availability and health of your cloud resources.

**What to Monitor?**

- CPU, Memory, Disk, Network
- Application Performance
- Error Rates & Latency
- Availability & Uptime
- Cost & Usage

**Monitoring Flow**




**Popular Tools**

| AWS | Azure | GCP |
|---|---|---|
| CloudWatch | Azure Monitor | Cloud Monitoring |
| CloudTrail | Activity Log | Cloud Audit Logs |
| X-Ray | Application Insights | Trace |
| SNS | Alerts | Alerting |

**Pro Tip**

Set smart alerts (not too many, not too few). Right alerts = Faster response & less downtime.

# 15) Cloud Cost Optimization

Optimize cost without compromising performance.

**Right Sizing**
Use right instance size. Don't over provision.

**Auto Scaling**
Scale up/down automatically based on demand.

**Reserved Instances / Savings Plans**
Commit for 1 or 3 years and save up to 70%.

**Spot Instances**
Use spare capacity at lower prices. Good for batch jobs.

**Storage Lifecycle**
Move data to cheaper storage tiers.

**Stop Unused Resources**
Stop or terminate resources when not in use.

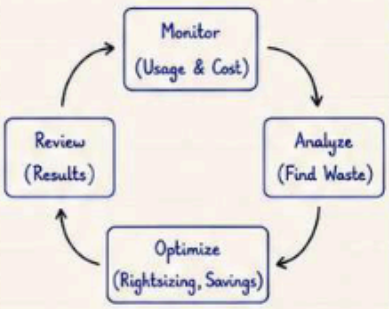
**Delete Unused Snapshots & IPs**
Clean up old snapshots, EBS, IPs regularly.

**Use Serverless Where Possible**
Pay only for what you use.

**Monitor & Analyze Cost**
Use cost explorer and set budgets & alerts.

**Tagging & Cost Allocation**
Use tags to track and allocate costs properly.

**Cost Optimization Cycle**




**Pro Tip**

Small daily optimizations can lead to huge savings at the end of the month!